Michigan Tech

Manifold Coordinates and Return Map Derivatives

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MA4535 — Nonlinear Dynamics & Chaos | Topic notes | September 27, 2026

Contents

Surface Coordinates and Return-Time Sensitivities

Nearby Starts Return at Different Times

- Think of the section Σ as a checkpoint. Start two trajectories at nearby points on it and follow each to its next crossing in the chosen direction.
- We want to know how a small change in the starting point changes the next crossing point. That is the sensitivity of the return map.
- When the first trajectory reaches the section checkpoint, the second may still be approaching it or may already have passed the checkpoint.
- A snapshot at that instant mixes two effects: where the trajectories return and when they return. Both need to reach the checkpoint.

Why the Sensitivity Needs a Correction

- The usual sensitivity calculation compares trajectories after the same amount of time.
- To compare crossing points, account for the second trajectory's extra or missing travel time. The correction follows the motion near the crossing, not the shortest path to the surface.
- The section is curved. Changing either of its two coordinates can change all three space coordinates; dropping one component loses part of the change.
- A trajectory that nearly skims the section makes this delicate.

Correcting Sensitivities with Calculus

- This is all really just the multivariable chain rule!
- Parameterize the surface and differentiate to find two tangent vectors.
- Follow a change in each parameter through the motion.
- Differentiate the surface equation to find the change in return time.
- Correct the returned point, then read off its two coordinates.
- After the calculation, organize the same ideas using maps and matrices.

Rössler System

- Fix parameters a, b, c with $b \neq 0$. The state is $\boldsymbol{y} \in \mathbb{R}^3$.

$$\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y}), \qquad \boldsymbol{f}(\boldsymbol{y}) = \begin{bmatrix} -y_2 - y_3 \\ y_1 + a y_2 \\ b + y_3(y_1 - c) \end{bmatrix}$$

- Write the solution starting at $\boldsymbol{q}$ as $\boldsymbol{y}(t; \boldsymbol{q})$. The **Flow** is the family of time maps

$$\phi_t(\boldsymbol{q}) = \boldsymbol{y}(t; \boldsymbol{q}), \qquad \phi_0(\boldsymbol{q}) = \boldsymbol{q}$$

- Each time map takes a starting point in three-dimensional space to its position after time t, where the solution exists.

Plane Section $y_1 = c$

- First use the plane with the same parameter c as the system.

$$h_{\mathrm{pl}}(\boldsymbol{y}) = y_1 - c, \qquad \Sigma_{\mathrm{pl}}^{-} = \{(c, u, v) : u + v > 0\}$$

  Here $Dh_{\mathrm{pl}}\boldsymbol{f} = -u - v < 0$: keep negative crossings.

- Coordinates and their tangent vectors are

$$\psi_{\mathrm{pl}}(u, v) = (c, u, v), \qquad \boldsymbol{B} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad \boldsymbol{S} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Start at $\boldsymbol{q}_1 = (c, u_1, v_1)$ and assume a next negative crossing $\boldsymbol{q}_2 = (c, u_2, v_2)$ exists at time τ. This plane is a different section from the top-of-the-hill surface.

Plane Section Return-Time Correction

- Integrate the state and its sensitivity together:

$$\boldsymbol{J}' = D\boldsymbol{f}(\boldsymbol{y}(t))\boldsymbol{J}, \qquad \boldsymbol{J}(0) = \boldsymbol{I}_3, \qquad \boldsymbol{J} = \boldsymbol{J}(\tau; \boldsymbol{q}_1)$$

- At the returned point, write

$$\boldsymbol{f}_2 = \boldsymbol{f}(\boldsymbol{q}_2) = (F_1, F_2, F_3) = (-u_2 - v_2,\ c + au_2,\ b)$$

- A starting change is $\delta\boldsymbol{q}_1 = (0, \delta u, \delta v)$. The returned change must have first component zero:

$$0 = J_{12}\delta u + J_{13}\delta v + F_1\delta\tau, \qquad \delta\tau = -\frac{J_{12}\delta u + J_{13}\delta v}{F_1}$$

  The chosen crossing direction gives $F_1 < 0$.

Project Along the Flow onto the Plane

- Account for the change in return time:

$$\delta \boldsymbol{q}_2 = \boldsymbol{J}\delta \boldsymbol{q}_1 + \boldsymbol{f}_2 \delta\tau = \boldsymbol{P}_2 \boldsymbol{J}\delta \boldsymbol{q}_1$$

- With $\boldsymbol{N} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$,

$$\boldsymbol{P}_2 = \boldsymbol{I}_3 - \frac{\boldsymbol{f}_2 \boldsymbol{N}}{F_1} = \begin{bmatrix} 0 & 0 & 0 \\ -F_2/F_1 & 1 & 0 \\ -F_3/F_1 & 0 & 1 \end{bmatrix}$$

- $\boldsymbol{N}\boldsymbol{P}_2 = 0$ puts the change in the plane, and $\boldsymbol{P}_2\boldsymbol{f}_2 = \boldsymbol{0}$ removes the flow direction. This projection follows the flow; it is generally not orthogonal.

Plane Section Return Derivative

- Lift the initial coordinate change, propagate it, correct the return time, and extract the final coordinates:

$$\boldsymbol{K} = \boldsymbol{S}\boldsymbol{P}_2\boldsymbol{J}\boldsymbol{B} = \begin{bmatrix} J_{22} - \dfrac{F_2}{F_1}J_{12} & J_{23} - \dfrac{F_2}{F_1}J_{13} \\ J_{32} - \dfrac{F_3}{F_1}J_{12} & J_{33} - \dfrac{F_3}{F_1}J_{13} \end{bmatrix}$$

- Thus $\delta\boldsymbol{u}_2 = \boldsymbol{K}\delta\boldsymbol{u}_1$ to first order, with $\boldsymbol{u}_j = (u_j, v_j)$.
- Taking only the lower-right block of $\boldsymbol{J}$ misses the return-time correction. The curved section uses the same steps, with a different normal and different coordinate tangent vectors.

Top of the Hill Section

- A peak in $y_3(t)$ must satisfy $y_3' = 0$. These points lie on the surface

$$h(\boldsymbol{y}) = b + y_3(y_1 - c), \qquad \Sigma = \{\boldsymbol{y} \in \mathbb{R}^3 : h(\boldsymbol{y}) = 0\}$$

- The gradient is perpendicular to the surface. Along a trajectory, the chain rule gives

$$\nabla h(\boldsymbol{y}) = (y_3, 0, y_1 - c), \qquad y_3'' = \nabla h(\boldsymbol{y}) \cdot \boldsymbol{f}(\boldsymbol{y})$$

- Choose the crossings at strict local maxima, where $y_3'' < 0$.

$$\Sigma^- = \{\boldsymbol{q} \in \Sigma : \nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) < 0\} \subset \Sigma \subset \mathbb{R}^3$$

The opposite sign defines Σ^+. Zero excludes a transverse crossing.

Surface Points and Two Coordinates

- Use $\boldsymbol{u} = (u, v) = (y_2, y_3) \in \mathbb{R}^2$. Since $b \neq 0$, every point on Σ has $v \neq 0$ and $y_1 = c - b/v$.
- The **Parameterization** and its coordinate domain are

$$U = \{(u, v) \in \mathbb{R}^2 : v \neq 0,\ -v(u + v) < 0\}$$

$$\boldsymbol{\psi} : U \to \Sigma^- \subset \mathbb{R}^3, \qquad \boldsymbol{\psi}(u, v) = \begin{bmatrix} c - b/v \\ u \\ v \end{bmatrix}$$

- Indeed, $h(\boldsymbol{\psi}(u, v)) = 0$ and $\nabla h(\boldsymbol{\psi}(u, v)) \cdot \boldsymbol{f}(\boldsymbol{\psi}(u, v)) = -v(u + v) < 0$.

Two Tangent Vectors from Two Parameters

- Differentiate $\psi(u, v) = (c - b/v, u, v)$ one parameter at a time. Call the resulting **Coordinate Tangent Vectors**

$$\boldsymbol{g}_u = \frac{\partial \psi}{\partial u} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \boldsymbol{g}_v = \frac{\partial \psi}{\partial v} = \begin{bmatrix} b/v^2 \\ 0 \\ 1 \end{bmatrix}$$

- Keep their lengths: these vectors measure change per unit of parameter. They are not normalized; $\boldsymbol{g}_u$ happens to have length one.
- For a curve $(u(s), v(s))$ in the parameter domain, Calc III gives

$$\frac{d}{ds}\psi(u(s), v(s)) = \boldsymbol{g}_u \frac{du}{ds} + \boldsymbol{g}_v \frac{dv}{ds} \in \mathbb{R}^3$$

Every small motion along the surface is a combination of these two vectors.

Sensitivity at a Fixed Time

- The 3×3 sensitivity matrix records how each solution component changes with each starting coordinate, holding t fixed.

$$J_{ij}(t; \boldsymbol{q}) = \frac{\partial y_i(t; \boldsymbol{q})}{\partial q_j}, \qquad \delta \boldsymbol{y} \approx \boldsymbol{J} \delta \boldsymbol{q}$$

- Integrate this matrix together with the state.

$$\boldsymbol{J}'(t) = \begin{bmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ y_3(t) & 0 & y_1(t) - c \end{bmatrix} \boldsymbol{J}(t), \qquad \boldsymbol{J}(0) = \boldsymbol{I}_3$$

- For a change in $\alpha = u$ or v, the starting change is $\boldsymbol{g}_\alpha \, \delta\alpha$. At fixed time, the resulting change is approximately $\boldsymbol{J}\boldsymbol{g}_\alpha \, \delta\alpha$.

Forward Map and the Calc III Chain Rule

- Start at $\boldsymbol{q}_1 = \psi(u, v)$, with $(u, v) \in U$. Assume a smooth next return to Σ^-. Let $T(u, v)$ be its time.

$$T(u, v) > 0, \qquad \boldsymbol{r}(u, v) = \phi_{T(u,v)}(\psi(u, v)) = \boldsymbol{q}_2 \in \Sigma^- \subset \mathbb{R}^3$$

- Both the starting point and the travel time change. For $\alpha = u, v$,

$$\boldsymbol{r}_\alpha = \underbrace{\boldsymbol{J}\boldsymbol{g}_\alpha}_{\text{change at fixed time}} + \underbrace{\boldsymbol{f}(\boldsymbol{q}_2)T_\alpha}_{\text{extra travel}}$$

  Subscripts mean partial derivatives. The Jacobian $\boldsymbol{J} = \boldsymbol{J}(T(u, v); \boldsymbol{q}_1)$ is evaluated at the return time.

- Read $(u_2, v_2) = (r_2, r_3)$ from the returned point. The chain rule gives

$$\boldsymbol{r}_\alpha = \boldsymbol{g}_u(u_2, v_2)\frac{\partial u_2}{\partial \alpha} + \boldsymbol{g}_v(u_2, v_2)\frac{\partial v_2}{\partial \alpha}$$

Return-Time Sensitivity by Implicit Differentiation

- Every returned point lies on the surface: $h(\boldsymbol{r}(u,v)) = 0$. Differentiate this scalar equation, just as in Calc III, for $\alpha = u, v$.

$$0 = \nabla h(\boldsymbol{q}_2) \cdot \boldsymbol{r}_\alpha = \nabla h(\boldsymbol{q}_2) \cdot \left(\boldsymbol{J}\boldsymbol{g}_\alpha + \boldsymbol{f}(\boldsymbol{q}_2)\, T_\alpha \right)$$

- Solve one scalar equation for the extra travel time per unit of parameter.

$$T_\alpha = -\frac{\nabla h(\boldsymbol{q}_2) \cdot \boldsymbol{J}\boldsymbol{g}_\alpha}{\nabla h(\boldsymbol{q}_2) \cdot \boldsymbol{f}(\boldsymbol{q}_2)}$$

  The denominator is nonzero at a transverse crossing.
- Substitute back into $\boldsymbol{r}_\alpha$ to get the corrected tangent vector. Define $\widehat{\boldsymbol{p}}(u,v) = (r_2(u,v), r_3(u,v))$; its sensitivities are

$$D\widehat{\boldsymbol{p}}(u,v) = \begin{bmatrix} (\boldsymbol{r}_u)_2 & (\boldsymbol{r}_v)_2 \\ (\boldsymbol{r}_u)_3 & (\boldsymbol{r}_v)_3 \end{bmatrix}, \qquad \delta T \approx T_u\, \delta u + T_v\, \delta v$$

Return Map on the Section

- We now describe the same calculation as maps between spaces. Let $V \subset \Sigma^-$ be an open set with a smooth first return to Σ^-. The **Return Time** is

$$\tau : V \to (0, \infty), \qquad T(u, v) = \tau(\psi(u, v))$$

- The **Return Map** acts on the two-dimensional section.

$$\boldsymbol{p} : V \subset \Sigma^- \to \Sigma^- \subset \mathbb{R}^3, \qquad \boldsymbol{q}_1 \longmapsto \boldsymbol{q}_2 = \boldsymbol{p}(\boldsymbol{q}_1)$$

- Make the inclusion $\boldsymbol{\iota} : \Sigma^- \hookrightarrow \mathbb{R}^3$ explicit: $\boldsymbol{\iota}(\boldsymbol{q}) = \boldsymbol{q}$.

$$(\boldsymbol{\iota} \circ \boldsymbol{p})(\boldsymbol{q}_1) = \phi_{\tau(\boldsymbol{q}_1)}(\boldsymbol{\iota}(\boldsymbol{q}_1))$$

Ambient Maps and the Return Map

- At $t_1 = \tau(\boldsymbol{q}_1)$, the fixed-time map $\phi_{t_1} : \Omega_{t_1} \subseteq \mathbb{R}^3 \to \mathbb{R}^3$ agrees with $\boldsymbol{p}$ at $\boldsymbol{q}_1$, but generally not nearby on V. Here Ω_{t_1} is the set where the solution exists through t_1.
- Transversality gives a smooth event-time branch $\widetilde{\tau} : N \subset \mathbb{R}^3 \to \mathbb{R}$ near $\boldsymbol{q}_1$, selecting the crossing near t_1.

$$h(\phi_{\widetilde{\tau}(\boldsymbol{y})}(\boldsymbol{y})) = 0, \qquad \widetilde{\tau}|_{N\cap V} = \tau|_{N\cap V}$$

- The ambient event map is

$$\boldsymbol{e} : N \to \mathbb{R}^3, \qquad \boldsymbol{e}(\boldsymbol{y}) = \phi_{\widetilde{\tau}(\boldsymbol{y})}(\boldsymbol{y}), \qquad \boldsymbol{e}(N) \subset \Sigma^-$$

On $N \cap V$, $\boldsymbol{e} = \boldsymbol{\iota} \circ \boldsymbol{p}$; the domains differ.

Coordinates from a Surface Point

- The inverse chart is a map on the section.

$$\chi : \Sigma^- \to U \subset \mathbb{R}^2, \qquad \chi(\boldsymbol{q}) = (q_2, q_3), \qquad \boldsymbol{q} = (q_1, q_2, q_3)$$

  Here q_i denotes a component; $\boldsymbol{q}_1, \boldsymbol{q}_2$ denote crossings.
- The inverse identities hold on the stated spaces.

$$\chi \circ \psi = \mathrm{id}_U, \qquad \psi \circ \chi = \mathrm{id}_{\Sigma^-}$$

- The ambient projection $\pi : \mathbb{R}^3 \to \mathbb{R}^2$, $\pi(\boldsymbol{y}) = (y_2, y_3)$ has the same formula, but a larger domain.

$$\chi = \pi \circ \iota, \qquad \iota \circ \psi \circ \pi(\boldsymbol{y}) = (c - b/y_3, y_2, y_3)$$

  The last expression equals $\boldsymbol{y}$ only on the section, where defined.

Return Map in Coordinates

- Set $U_V = \chi(V) \subset U \subset \mathbb{R}^2$. The coordinate map is

$$\widehat{\boldsymbol{p}} = \boldsymbol{\chi} \circ \boldsymbol{p} \circ \psi : U_V \to U, \qquad \boldsymbol{u}_1 \longmapsto \boldsymbol{u}_2$$

- The two descriptions commute.

$$\begin{array}{ccc} \boldsymbol{u}_1 \in U_V \subset \mathbb{R}^2 & \xrightarrow{\widehat{\boldsymbol{p}}} & \boldsymbol{u}_2 \in U \subset \mathbb{R}^2 \\ \psi \downarrow \quad \uparrow \boldsymbol{\chi} & & \psi \downarrow \quad \uparrow \boldsymbol{\chi} \\ \boldsymbol{q}_1 \in V \subset \Sigma^- & \xrightarrow{\boldsymbol{p}} & \boldsymbol{q}_2 \in \Sigma^- \subset \mathbb{R}^3 \end{array}$$

- Explicitly, $\boldsymbol{q}_j = \psi(\boldsymbol{u}_j)$ and $\boldsymbol{u}_j = \boldsymbol{\chi}(\boldsymbol{q}_j)$ for $j = 1, 2$, with

$$\widehat{\boldsymbol{p}}(\boldsymbol{u}_1) = \boldsymbol{\pi}\big(\phi_{\tau(\psi(\boldsymbol{u}_1))}(\psi(\boldsymbol{u}_1))\big)$$

Tangent Spaces and Coordinate Derivatives

- Write $Dh = (\nabla h)^{\mathsf{T}}$ as a row. At $\boldsymbol{q} = \psi(u, v)$, the tangent plane contains the surface variations

$$T_{\boldsymbol{q}}\Sigma^{-} = \ker Dh(\boldsymbol{q}) \subset \mathbb{R}^3, \qquad v\,\delta y_1 + (q_1 - c)\delta y_3 = 0$$

- Collect the two coordinate tangent vectors as columns of $\boldsymbol{B}$.

$$\boldsymbol{B}(u, v) = D(\iota \circ \psi)(u, v) = \begin{bmatrix} 0 & b/v^2 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} : \mathbb{R}^2 \to T_{\boldsymbol{q}}\Sigma^{-} \subset \mathbb{R}^3$$

- With $\boldsymbol{S} = D\pi = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $D\chi(\boldsymbol{q}) = \boldsymbol{S}|_{T_{\boldsymbol{q}}\Sigma^{-}}$.

$$\boldsymbol{SB} = \boldsymbol{I}_2, \qquad (\boldsymbol{BS})|_{T_{\boldsymbol{q}}\Sigma^{-}} = \mathrm{id}_{T_{\boldsymbol{q}}\Sigma^{-}}$$

Return-Time Variation

- Write $\boldsymbol{J} = \boldsymbol{J}(\tau(\boldsymbol{q}_1); \boldsymbol{q}_1)$, $\boldsymbol{f}_2 = \boldsymbol{f}(\boldsymbol{q}_2)$, and $\boldsymbol{N}_2 = Dh(\boldsymbol{q}_2)$.
- For $\delta\boldsymbol{q}_1 \in T_{\boldsymbol{q}_1}\Sigma^-$, differentiation gives

$$\delta\boldsymbol{q}_2 = \boldsymbol{J}\delta\boldsymbol{q}_1 + \boldsymbol{f}_2\delta\tau \in T_{\boldsymbol{q}_2}\Sigma^-, \qquad \boldsymbol{N}_2\delta\boldsymbol{q}_2 = 0$$

- Solve the tangency constraint for the time variation.

$$\delta\tau = -\frac{\boldsymbol{N}_2\boldsymbol{J}\delta\boldsymbol{q}_1}{\boldsymbol{N}_2\boldsymbol{f}_2}, \qquad D\tau(\boldsymbol{q}_1) : T_{\boldsymbol{q}_1}\Sigma^- \to \mathbb{R}$$

The denominator is negative on Σ^-, so it is nonzero.

Ambient Projection and Section Derivative

- Project ambient variations along the flow onto the final tangent plane.

$$\boldsymbol{P}_2 = \boldsymbol{I}_3 - \frac{\boldsymbol{f}_2 \boldsymbol{N}_2}{\boldsymbol{N}_2 \boldsymbol{f}_2} : \mathbb{R}^3 \to T_{\boldsymbol{q}_2} \Sigma^- \subset \mathbb{R}^3$$

- The ambient event map has the 3×3 derivative

$$D\boldsymbol{e}(\boldsymbol{q}_1) = \boldsymbol{P}_2 \boldsymbol{J} : \mathbb{R}^3 \to \mathbb{R}^3$$

- The section derivative has a two-dimensional domain and codomain.

$$D\boldsymbol{p}(\boldsymbol{q}_1) : T_{\boldsymbol{q}_1} \Sigma^- \to T_{\boldsymbol{q}_2} \Sigma^-$$

$$D\iota(\boldsymbol{q}_2) D\boldsymbol{p}(\boldsymbol{q}_1) = \boldsymbol{P}_2 \boldsymbol{J} D\iota(\boldsymbol{q}_1)$$

Explicit Projection for the Curved Section

- Write the final crossing as $\boldsymbol{q}_2 = (x, y, z)$ and set

$$F_1 = -y - z, \qquad F_2 = x + ay, \qquad \boldsymbol{N}_2 = \begin{bmatrix} z & 0 & x - c \end{bmatrix}$$

  On the section, $f_3(\boldsymbol{q}_2) = 0$ and $\boldsymbol{N}_2 \boldsymbol{f}_2 = zF_1 < 0$.

- The projection is

$$\boldsymbol{P}_2 = \begin{bmatrix} 0 & 0 & -(x-c)/z \\ -F_2/F_1 & 1 & -F_2(x-c)/(zF_1) \\ 0 & 0 & 1 \end{bmatrix}$$

- $\boldsymbol{N}_2 \boldsymbol{P}_2 = 0$ and $\boldsymbol{P}_2 \boldsymbol{f}_2 = \boldsymbol{0}$. Tangent variations satisfy $\delta x = (b/z^2)\delta z$; the first row is generally nonzero.

Two-Dimensional Return Derivative

- Let $\boldsymbol{u}_j = (u_j, v_j) = \chi(\boldsymbol{q}_j)$ for $j = 1, 2$. The chain rule gives

$$\boldsymbol{K} = D\widehat{\boldsymbol{p}}(\boldsymbol{u}_1) = \underbrace{\boldsymbol{S}}_{2\times 3}\underbrace{\boldsymbol{P}_2}_{3\times 3}\underbrace{\boldsymbol{J}}_{3\times 3}\underbrace{\boldsymbol{B}(\boldsymbol{u}_1)}_{3\times 2} : \mathbb{R}^2 \to \mathbb{R}^2$$

- The variation maps have explicit spaces.

$$\delta\boldsymbol{u}_1 \in \mathbb{R}^2 \xrightarrow{\boldsymbol{B}(\boldsymbol{u}_1)} \delta\boldsymbol{q}_1 \in T_{\boldsymbol{q}_1}\Sigma^- \xrightarrow{D\boldsymbol{p}(\boldsymbol{q}_1)} \delta\boldsymbol{q}_2 \in T_{\boldsymbol{q}_2}\Sigma^- \xrightarrow{\boldsymbol{S}} \delta\boldsymbol{u}_2 \in \mathbb{R}^2$$

- Thus $\delta\boldsymbol{u}_2 = \boldsymbol{K}\delta\boldsymbol{u}_1$ and $\delta\boldsymbol{q}_j = \boldsymbol{B}(\boldsymbol{u}_j)\delta\boldsymbol{u}_j$. Deleting a row and column of $\boldsymbol{P}_2\boldsymbol{J}$ would omit the initial lift.

Explicit Coordinate Jacobian

- With $\beta = b/v_1^2$, form the 3×2 matrix

$$\boldsymbol{H} = \boldsymbol{J}\boldsymbol{B}(\boldsymbol{u}_1), \qquad H_{i1} = J_{i2}, \quad H_{i2} = J_{i3} + \beta J_{i1}, \quad i = 1, 2, 3$$

- At $\boldsymbol{q}_2 = (x, y, z)$, set $r = F_2/F_1$ and $k = (x - c)/z$. Then

$$\boldsymbol{K} = \begin{bmatrix} H_{21} - rH_{11} - rkH_{31} & H_{22} - rH_{12} - rkH_{32} \\ H_{31} & H_{32} \end{bmatrix}$$

- The integrator supplies $\tau(\boldsymbol{q}_1)$, $\boldsymbol{q}_2$, and $\boldsymbol{J}$. The chart supplies $\boldsymbol{B}(\boldsymbol{u}_1)$ and $\boldsymbol{S}$; the final crossing supplies $\boldsymbol{P}_2$.

References I